

Calculus (PR12)

Week 7: Multivariable Optimization

Learning Goals



Critical Points of Functions of two variables

- **Compute** and **interpret** partial derivatives.
- Use the gradient to find **locate candidate maxima, minima, and saddle points.**



Classify Critical Points

- Interpret the **Hessian matrix** as a measure of **local curvature.**
- **Distinguish** between local minima, maxima, and saddle points using the **second derivative tests.**



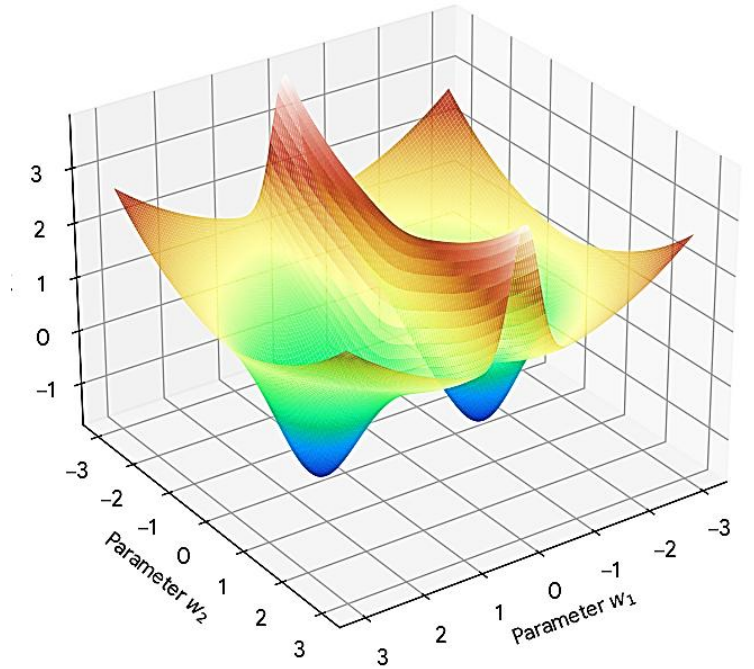
Constraint Optimization Problems

- **Explain** the geometric idea behind **Lagrange multipliers.**
- **Apply the Lagrange Multipliers Method** to find constrained optima.

Optimization is Everywhere in AI

Many AI tasks are optimization problems:

- Training neural networks
- Minimizing prediction error
- Finding best model parameters
- Resource allocation and decision making



Question:

How do we know whether a point is a minimum, maximum, or neither?

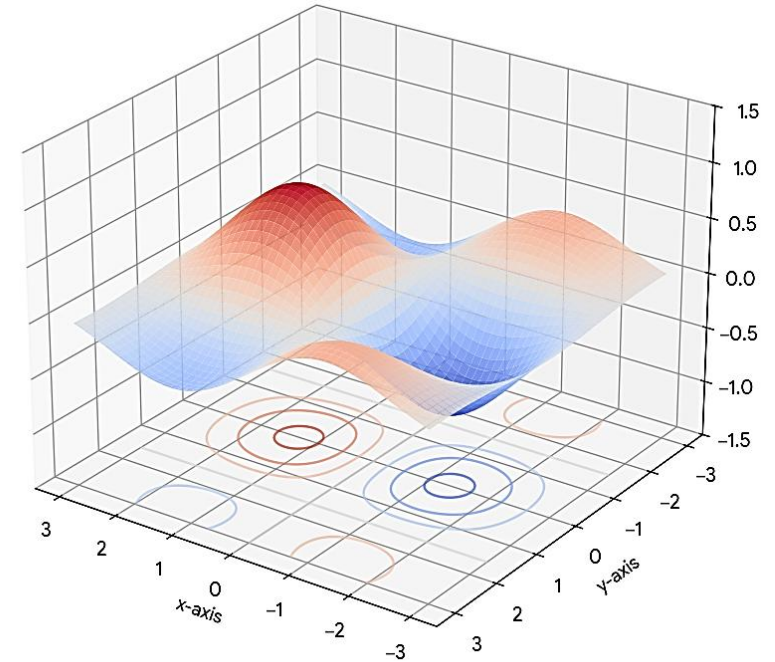
Recall from Week 3

For a function $f(x)$:

- Critical point $f'(x) = 0$
- Second derivative indicates the shape

Now we move to:

$$z = f(x, y)$$



Examples:

Temperature on a map, cost depending on two parameters, AI loss depending on two weights

Optimization

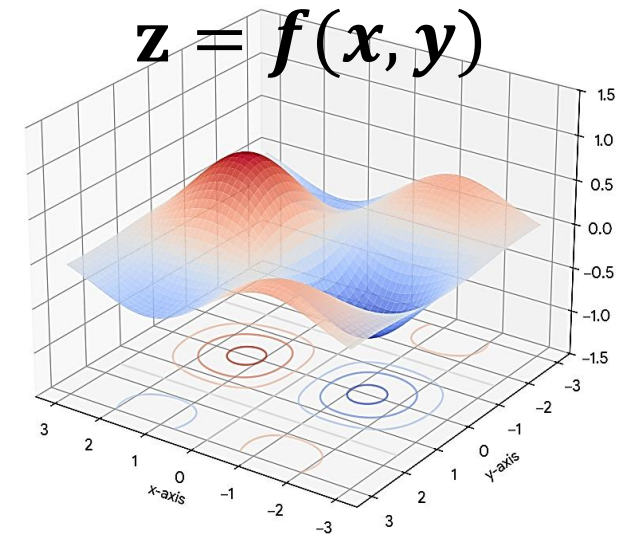
A **CRITICAL POINT** is where:

$$\frac{\partial z}{\partial x} = 0 \text{ and } \frac{\partial z}{\partial y} = 0$$

Meaning:

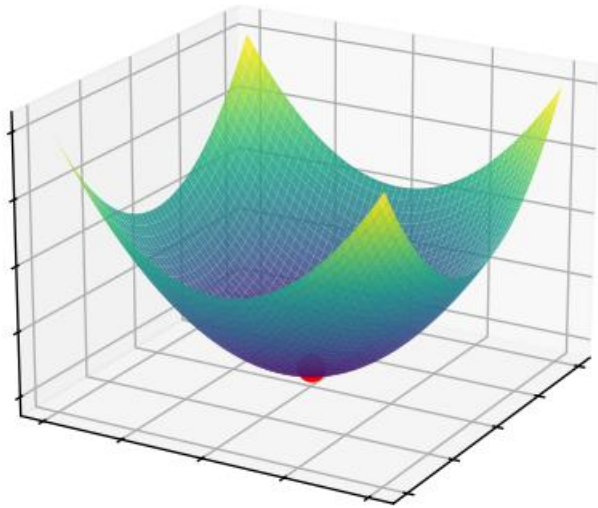
- No local change in the x - direction.
- No local change in the y - direction.

Interpretation: The surface is momentarily flat in every direction.

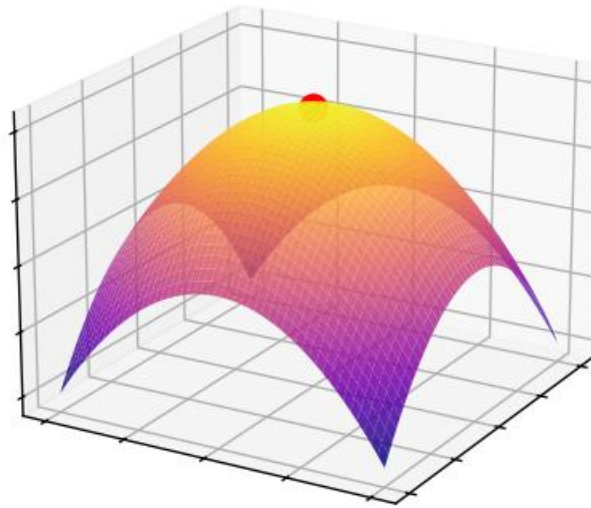


Critical Points

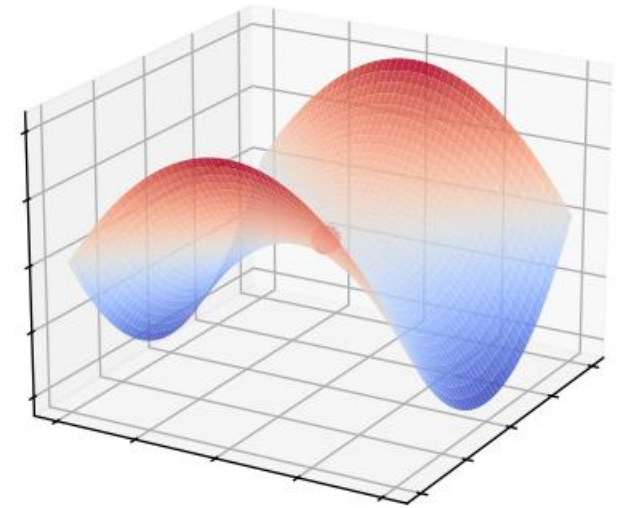
Where are $\frac{\partial z}{\partial x} = 0$ and $\frac{\partial z}{\partial y} = 0$?



Local Minimum



Local Maximum



Saddle Point

Critical Points in Machine Learning

What are critical points in Machine Learning?

$$Loss = L(w_1, w_2, \dots)$$

Critical points occur when gradients vanish:

$$\nabla L = 0$$

Questions:

- 1. Have we found a good solution?**
- 2. Is it a minimum?**
- 3. Is it a saddle?**



Need More Information

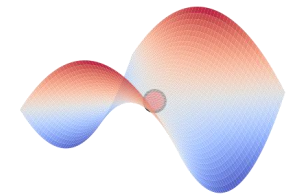
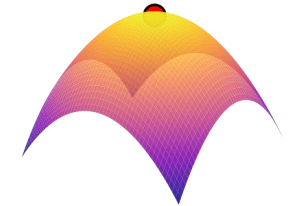
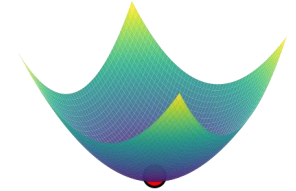
At a critical point:

$$f_x = f_y = 0$$

But this condition alone cannot tell us whether we have:

- Maximum
- Minimum
- Saddle

Therefore, we need more information about curvature.



Curvature and Second Derivatives

Measuring shape: Second derivatives describe how the surface bends.

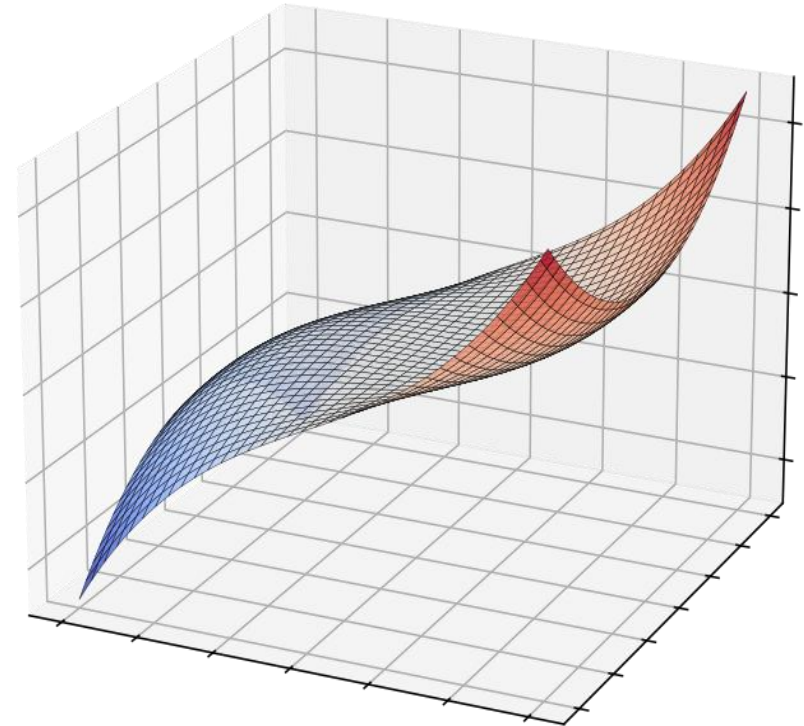
Important quantities:

$$f_{xx}$$

$$f_{yy}$$

$$f_{xy}$$

These capture curvature in different directions.



The Hessian Matrix

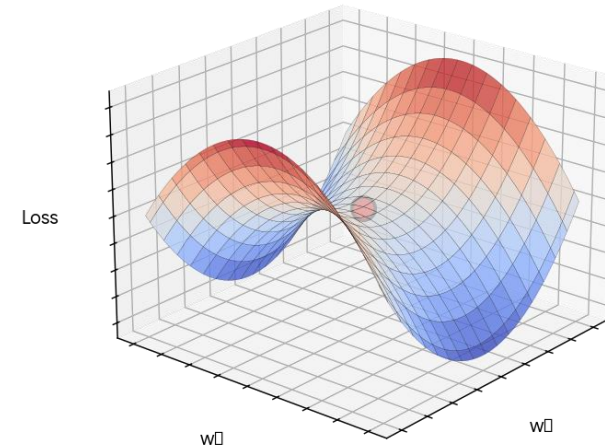
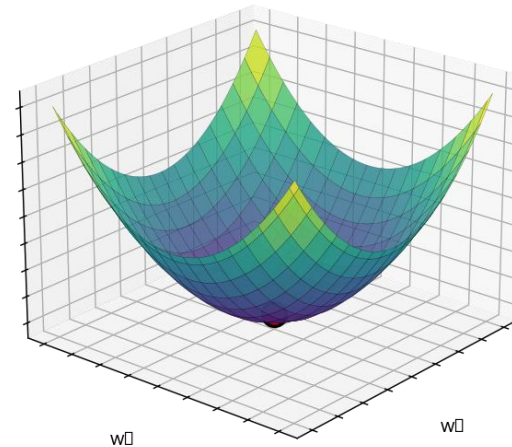
Organizing curvature information by Hessian matrix:

$$H = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}$$

Interpretation: The Hessian summarizes local curvature.

In AI:

- Hessian tells us about the shape of the loss landscape.
- Useful for advanced optimization methods.



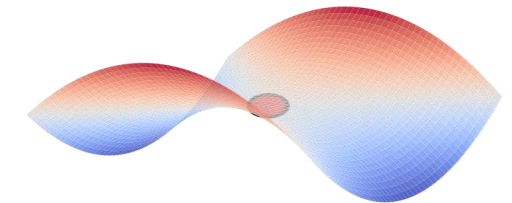
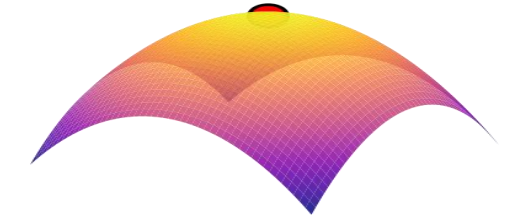
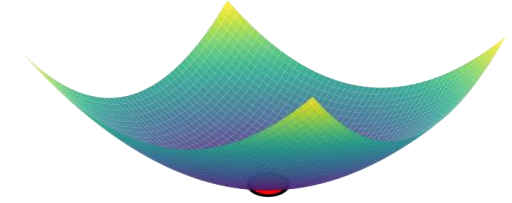
Hessian Test Intuition

At a critical point:

Hessian Positive Definite: *Local minimum*

Hessian Negative Definite: *Local maximum*

Hessian Indefinite: *Saddle point*



$$H = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} = \begin{bmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{bmatrix}$$

$$D(H) > 0 \text{ \& } H_{11} > 0$$

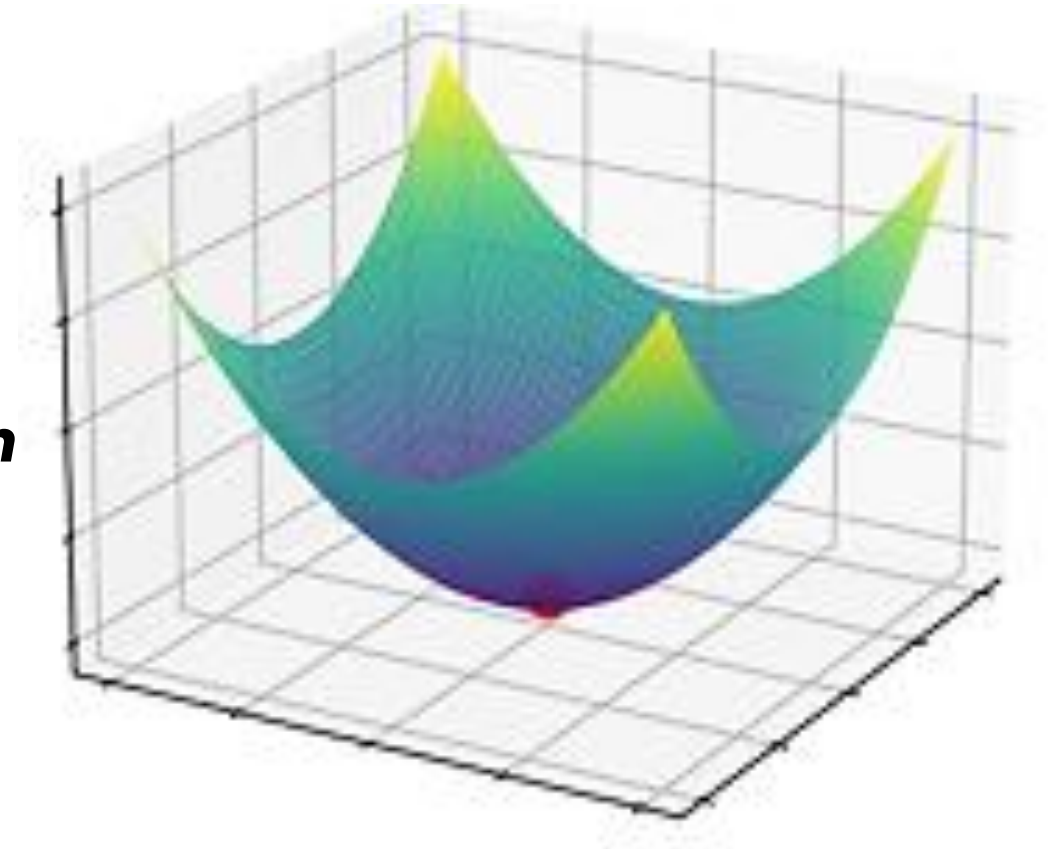
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$$D(H) > 0 \text{ \& } H_{11} < 0$$

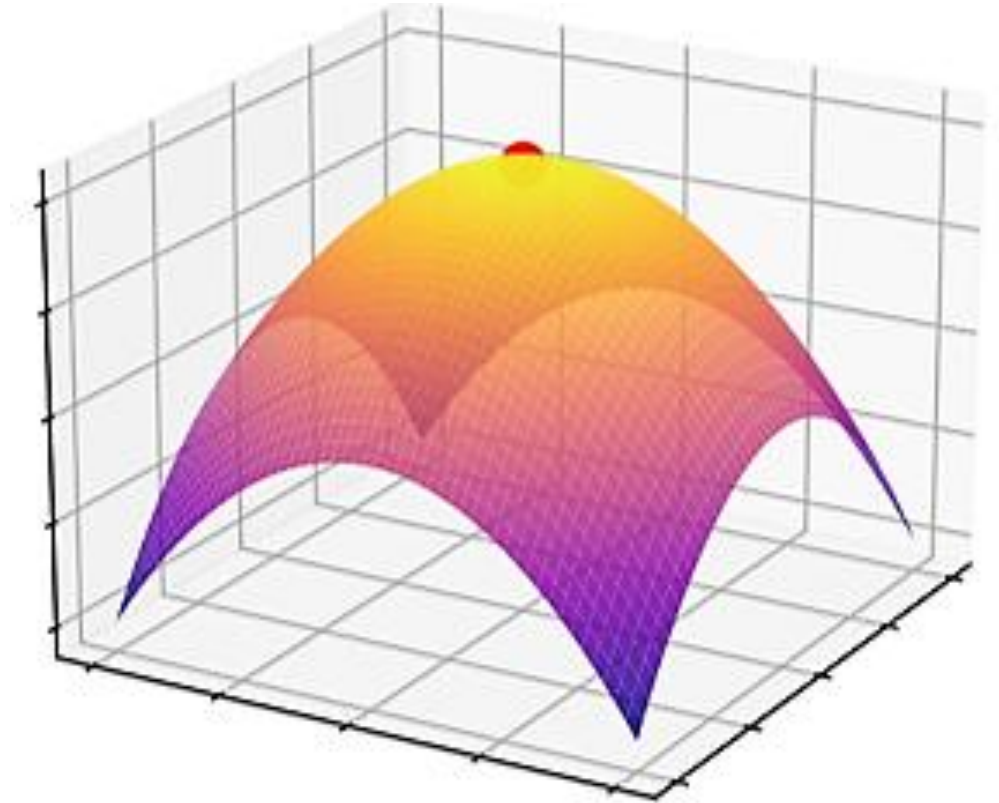
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$$D(H) < 0$$

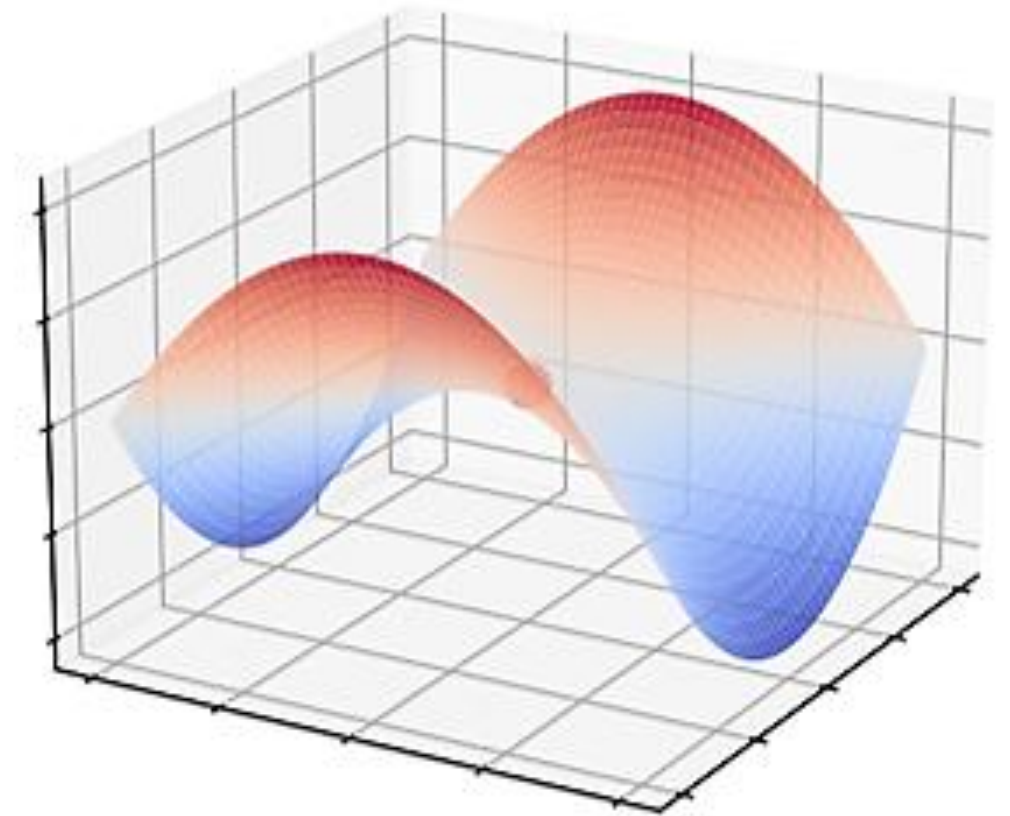
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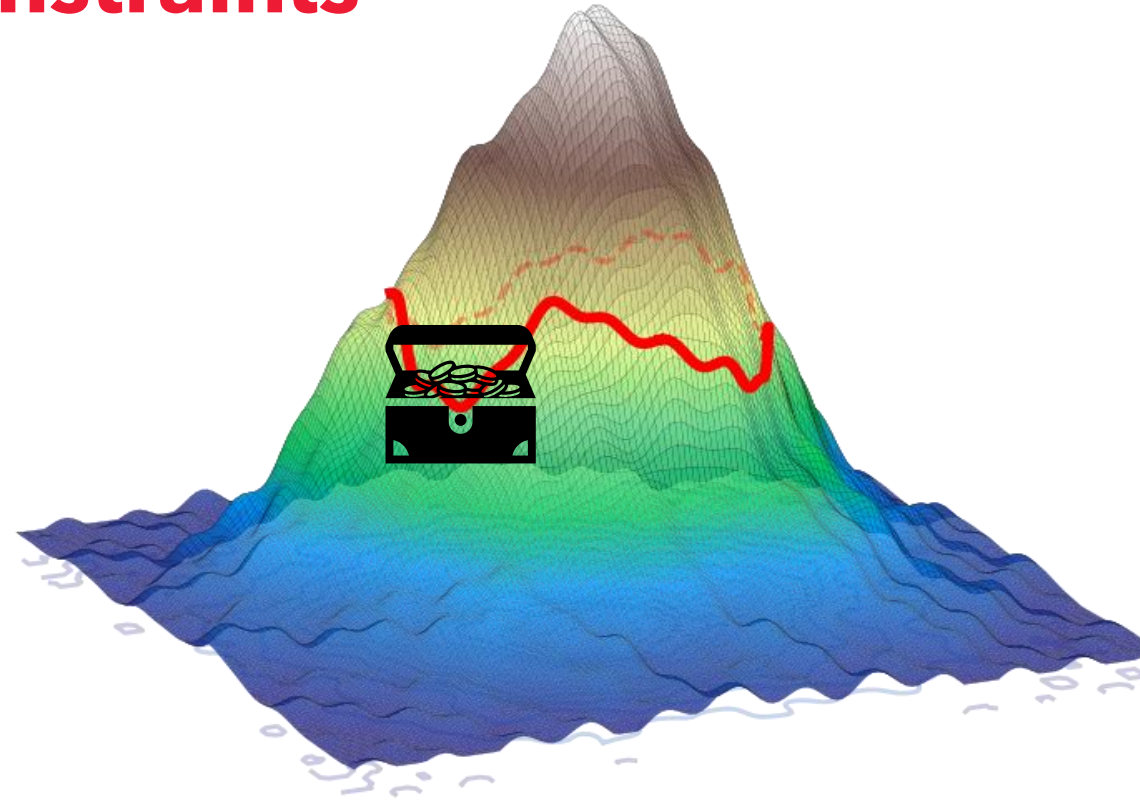
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Introducing Constraints

Real Problem have restrictions, for instance:

- Fixed budget
- Limited memory
- Energy constraints
- Probability distributions summing to 1

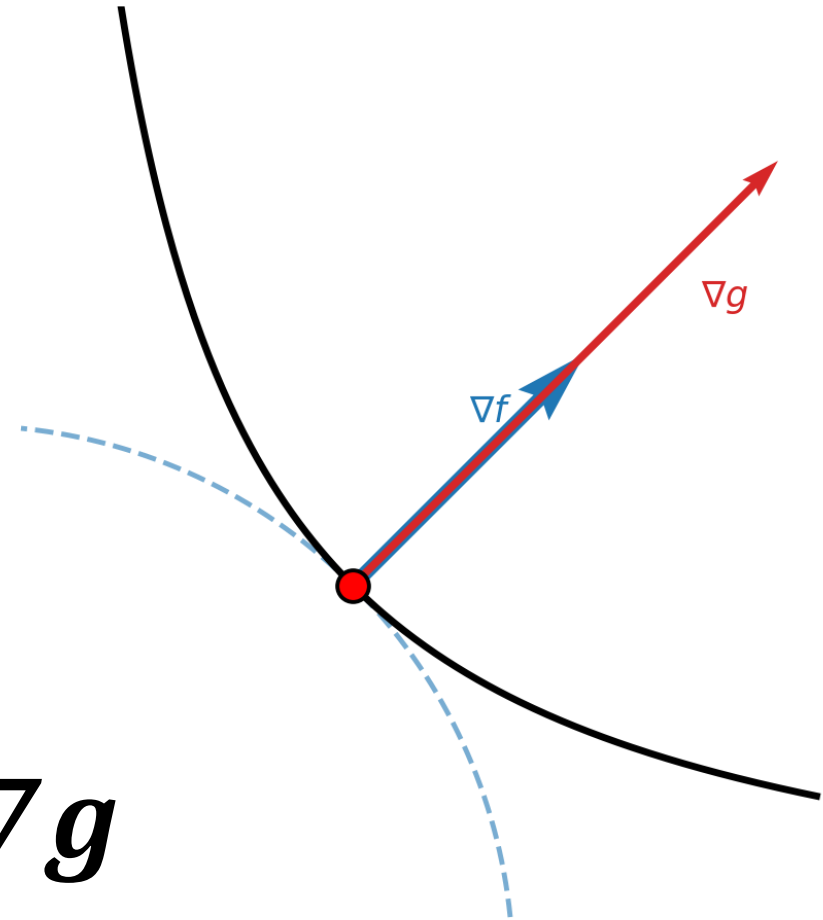


Now optimization becomes:

Optimize objective $f(x, y)$, subject to a constraint $g(x, y) = c$.

Core Idea of Lagrange Multipliers

What happens at the Optimum?
At the constrained optimum, objective cannot improve while staying on the constraint, meaning ∇f and ∇g point in the same direction.



$$\nabla f = \lambda \nabla g$$

The Lagrange Multiplier Equation

To solve **"optimize $f(x, y)$, subject to $g(x, y) = c$ "**:

Solve:

$$\nabla f = \lambda \nabla g$$

Together with :

$$g(x, y) = c$$

Takeaways

Key Ideas:

- Critical points occur when the gradient is zero.
- The Hessian describes local curvature.
- Positive definite Hessian $\rightarrow$ minimum.
- Negative definite Hessian $\rightarrow$ maximum.
- Indefinite Hessian $\rightarrow$ Saddle.
- Constraints require the new idea of **Lagrange Multipliers**.

Next week:

Recapitulation and Exam Preparation

?

Questions

- Follow-up reading: Calculus: A Complete Course, (sections 10.7 & 14.1 – 14.3)
- Joining the Discussion Groups and the Lab

